MIL-J-7399D(USAF)
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SUPERSEDING
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30 Mar 1967

MILITARY SPECIFICATION

JACK, HYDRAULIC, TRIPOD

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hydraulic tripod jack, hand operated, 5-ton capacity, designated type B-5.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

	PPP-B-601	Boxes, Wood, Cleated-Plywood.

MILITARY

	MIL-P-116	Preservation, Methods of.
	MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
*	MIL-H-81019	Hydraulic Fluid, Petroleum Base, Ultra Low Temperature.

STANDARDS

FEDERAL

*	FED-STD-595	Color

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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MILITARY

	MIL-STD-129	Marking for Shipment and Storage.
	MIL-STD-130	Identification Marking of U.S. Military Property.
	MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of.
*	MIL-STD-808	Finish, Protective and Codes for Finishing Scheme for Ground and Ground Support Equipment.
*	MIL-STD-810	Environmental Test Methods.
	MIL-STD-831	Test Reports, Preparation of.
	MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
	MIL-STD-1188	Commercial Packaging of Supplies and Equipment.

DRAWING

AIR FORCE

50K25177	Jack Assembly - Hand, Hydraulic, Folding Tripod, 5 Ton Capacity, Type B-5.
7545352	Requirements for finishes, Protective and codes for SA-ALC Ground and Ground Support Equipment.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. This specification makes provisions for first article testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 AN or MIL standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part number. Commercial utility parts such as screws, bolts, nuts, cotter pins, etcetera, may be used, provided they develop suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and if practicable, on the contractor's drawings. In case there is no suitable corresponding AN or MIL standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction. Design and construction of the jack shall conform to Drawing 50K25177. When the drawing and this specification conflict, this specification shall govern.

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3.4 Performance. The jack shall be capable of satisfactory operation when subjected to the tests in 4.6.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.6 Workmanship. The jack, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies.

* 3.7 Paint and treatment. Finish per MIL-STD-808 as supplemented by USAF drawing 7545352. Color shall be number 24052 per FED-STD-595.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the jack shall be classified as follows:

a. First article tests.

b. Acceptance tests.

* 4.3 Test conditions. The jack shall be serviced with hydraulic fluid conforming to MIL-H-5606 except for the low temperature test which shall be conducted using fluid conforming to MIL-H-81019.

* 4.4 First article testing.

* 4.4.1 Test sample. The first article test sample shall consist of one jack representative of the production equipment. It shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract or purchase order, at the contractor's plant under the supervision of the procuring activity (see 6.2).

* 4.4.2 Test report. Upon completion of first article testing, the contractor shall submit a test report prepared in accordance with MIL-STD-831. Three copies of the test report are required.

* 4.4.3 First article tests. The first article tests shall consist of all the tests in 4.6.

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4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling test.

4.5.1 Individual tests. Individual tests shall consist of all the tests in 4.6.1.

4.5.2 Sampling test. Five jacks which have been manufactured under essentially the same conditions shall be selected at random from every 50 or fraction thereof produced and subjected to the following test.

4.5.2.1 Staticproof. The jack shall be tested by raising a 7.5-ton load vertically from the lowest position to the fully extended position. At the fully extended position, under a 5-ton load, a horizontal load of 0.75 ton shall be applied to the top of the screw extension. Failure of any part of the jack shall be cause for rejection.

4.5.2.2 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all the sampling tests shall be repeated.

* 4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each jack shall be inspected to determine compliance with the requirements specified herein with respect to workmanship, marking, and compliance with the drawings.

4.6.1.1 Each ram of the assembled jack shall be tested by applying a minimum hydraulic pressure of 3,000 pounds per square inch. Any leaking of hydraulic fluid shall be cause for rejection.

4.6.1.2 Each pump shall be tested by sealing the pump outlet and operating the pump until the bypass valve opens. The bypass valve shall be adjusted as specified on the drawings.

* 4.6.2 Low temperature test. The jack shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502, Procedure 1. The jack assembly shall be placed in a test chamber and cooled to -54° Centigrade (C) for a period of 48 hours. At the end of this period, while at -54°C, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened, and the ram shall retract with not more than 50 pounds applied to the ram. For this test, the jack shall be serviced with hydraulic fluid conforming to MIL-H-81019. The time and number of full strokes necessary to fully extend the ram shall be recorded at both 21° and -54°C. The time and strokes required to fully extend the ram at -54°C shall be not more than double the time and strokes required at 21°C.

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4.6.3 Life test. The life test shall consist of 200 cycles of raising a 5-ton load vertically, and 50 cycles of raising a 5-ton load vertically with a 0.75-ton horizontal load applied to the top of the screw extension. One cycle shall consist of raising the ram from collapsed height to the fully extended position and returning it to the collapsed position. The horizontal load shall be graduated from zero in the collapsed position to 0.75 ton in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.7 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A, Level C or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The jack shall be disassembled to accomplish reduction in cubage. Disassembly shall be limited to the extent that reassembly can be performed without requiring the use of special tools or skills. Removed bolts, nuts, washers and pins shall be placed in one of the mating parts and secured to prevent loss.

5.1.1.2 Preservation. The jack shall be preserved in accordance with MIL-P-116, Method I. All bare unpainted metal surfaces subject to rust or corrosion shall be coated with type P2 preservative in accordance with MIL-P-116.

* 5.1.2 Level C. Preservation and packaging shall afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use.

* 5.1.3 Commercial. Each tester shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. Each jack preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-601, style A, overseas type. Containers shall be provided with three nominal 4 X 4 inch skids and shall extend the full exterior width of the box and shall meet the requirements of PPP-B-601. Containers shall be closed and strapped in accordance with the appendix of the specification.

5.2.2 Level B. Same requirements as Level A, except the containers shall be domestic type.

5.2.3 Level C. Jacks packaged as specified in 5.1.2 shall be packed in a manner which affords adequate protection, at the lowest rate, against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

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* 5.2.4 Commercial. Each tester shall be packed in accordance with MIL-STD-1188.

5.3 Physical protection. All cushioning, blocking, bracing and bolting shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The type B-5 jack covered by this specification is intended for use as an aircraft wing jack.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Site of first article testing (see 4.4.1).

c. Applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.4 Certain provisions of this specification are the subject of international standardization agreement ASCC Standard 11/7C and STANAG 3098. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-J-7399D(USAF)	JACK, HYDRAULIC, TRIPOD

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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